A

Report

on

"Machine Learning-Based Predictive Maintenance for Mining Equipment Using Multi-Sensor Data and Computer Vision"

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CAPSTONE PROJECT PROPOSAL

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ABSTRACT

This proposal presents a 12-week core AI/ML capstone project focused on predictive maintenance for mining equipment. The project will build machine-learning models from real industrial multi-sensor data and a separate computer-vision module for visible component surface defects. The primary sensor dataset will be TiNA, a public industrial anomaly dataset containing sensor, alarm, and maintenance-event data from a real ArcelorMittal mining machine. The computer-vision component will use a public industrial surface-defect image dataset to compare OpenCV-based feature extraction with CNN-based defect classification. The two modules are integrated at the decision level: the sensor model flags periods of elevated maintenance risk, and the vision module represents the inspection step that follows an alert. The final outcome will be a reproducible ML pipeline, validated model-comparison results, explainability outputs, weekly progress reports, and a small demonstration interface for equipment-health risk interpretation.

Index Terms—Anomaly detection, computer vision, explainable artificial intelligence, machine learning, mining equipment, predictive maintenance.

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ABBREVIATIONS

AI	Artificial Intelligence	EDA	Exploratory Data Analysis
ML	Machine Learning	SHAP	SHapley Additive exPlanations
PdM	Predictive Maintenance	PR-AUC	Precision-Recall Area Under the Curve
CV	Computer Vision	ROC-AUC	Receiver Operating Characteristic Area Under the Curve
CNN	Convolutional Neural Network	RUL	Remaining Useful Life

I. INTRODUCTION

A. Project Overview

Mining equipment operates under high mechanical load, vibration, dust, heat, long operating cycles, and harsh maintenance conditions. When heavy equipment fails unexpectedly, the result can include downtime, production loss, safety risk, and inefficient maintenance spending. This capstone proposes a practical ML-based predictive-maintenance system that uses industrial sensor histories and visual defect images to identify early warning signs of equipment degradation.

The project will keep a clear mining-equipment predictive-maintenance focus. TiNA will act as the primary sensor-data stream because it is based on a real ArcelorMittal mining machine and includes anonymized features, alarms, and maintenance codes. The computer-vision module will not replace the sensor model; it will be a secondary visual-inspection stream for industrial component surface defects using OpenCV and CNN-based methods.

Saskatchewan relevance is provided through the mining context: potash operations in the province are increasingly connected to automation, machine-health monitoring, sensors, predictive maintenance, and digital transformation. However, the project will not claim to use proprietary Saskatchewan mine telemetry. Instead, it will train on public real-world industrial datasets and frame the use case around mining equipment maintenance.

B. Problem Statement and Motivation

Reactive maintenance begins after a failure occurs, while fixed preventive maintenance may replace components too early or miss unexpected degradation. Mining equipment needs a more data-driven approach that can learn from sensor behaviour, maintenance events, alarms, and visible defects to provide earlier warning. The key problem is to develop and evaluate ML models that can identify maintenance-risk patterns without leaking future data or relying on unrealistic assumptions.

The project is relevant to industry because modern mining operations increasingly collect operational and condition-monitoring data from machines. It is also relevant to society because better maintenance can support safer operations, improved equipment availability, and more efficient use of resources. The academic motivation is to compare traditional ML, anomaly detection, and computer-vision approaches under a realistic 12-week postgraduate capstone scope.

C. Project Objectives and Research Questions

- Develop a reproducible preprocessing pipeline for large industrial multi-sensor time-series data.
- Construct leakage-safe predictive targets such as maintenance within a defined future time window.
- Engineer temporal, rolling, event-history, and anomaly-oriented sensor features.
- Compare baseline models with ensemble models such as Random Forest, XGBoost, and LightGBM.
- Use unsupervised anomaly detection to identify abnormal behaviour when labels are limited.
- Develop a secondary computer-vision pipeline using OpenCV features and CNN classification for visible industrial component defects.
- Use SHAP for sensor-model explanations and Grad-CAM for CNN visual explanations.

- Present weekly measurable progress to the instructor and complete the capstone project within 12 weeks.

Main research question: How can machine-learning methods applied to industrial sensor histories and visual component inspection support earlier, better-prioritized maintenance decisions for heavy mining equipment?

RQ1 (sensor module): How accurately can leakage-safe models trained on TiNA sensor, alarm, and maintenance-event data predict maintenance or alarm risk within a defined future time window?

RQ2 (vision module): How do OpenCV-based handcrafted features compare with CNN-based classification for detecting surface defects on industrial drive components?

RQ3 (secondary): Which feature representations and model families provide the best balance of failure-event recall, false-alarm control, interpretability, and feasibility within a 12-week capstone timeline?

D. Project Scope

In scope: sensor-based maintenance-risk prediction using TiNA; anomaly detection; OpenCV image preprocessing; classical ML and CNN-based visual defect classification; model evaluation; explainability; weekly progress reporting; final report and presentation.

Out of scope: live deployment in a mine; use of proprietary Saskatchewan mine telemetry; safety-certified decision-making; physical equipment integration; full Remaining Useful Life modelling as a mandatory deliverable. RUL will remain a stretch goal only if the selected data preparation and modelling progress are completed early.

II. RELATED WORK / BACKGROUND

Predictive maintenance combines condition monitoring, fault diagnosis, prognostics, and data-driven modelling. Recent reviews show that industrial PdM increasingly relies on sensor data, machine learning, deep learning, anomaly detection, and explainable methods to move from reactive maintenance to evidence-based maintenance decisions [1]–[9]. In industrial settings, evaluation is especially important because failure events are often rare, making accuracy alone a weak metric. Recall, F1-score, PR-AUC, false-negative rate, and maintenance cost trade-offs are more meaningful for predictive maintenance [10], [11]. Reviews focused on the mining sector report similar trends, together with sector-specific implementation challenges [12].

For mining relevance, public documents and industry reports indicate that Saskatchewan potash operations and the BHP Jansen project use high levels of automation, process sensors, machine-health monitoring, and equipment-health monitoring for predictive maintenance [13]–[15]. These sources support the practical relevance of the project without requiring access to proprietary mine telemetry.

TiNA is important because it contains time-series information from the sensor system of an actual ArcelorMittal mining machine. The dataset includes more than 38 million samples, 108 features, maintenance codes, maintenance subtypes, and alarms, while anonymizing sensor names and normalizing numeric values to protect industrial confidentiality [16]. This makes TiNA realistic but challenging: the project must design targets and features carefully instead of relying on obvious sensor names.

The Scania Component X and Scania APS datasets are included as supporting references and possible fallback/validation datasets because they represent real heavy-duty vehicle predictive-maintenance problems with operational readouts and repair or failure labels [17]–[19]. They are not mining datasets, but their heavy-equipment context is relevant to mobile mining equipment maintenance. The image dataset supports the visual inspection component by providing real industrial surface-defect images for OpenCV and CNN experimentation [20], [21]. Surface pitting is a rolling-contact fatigue failure mode

that also affects the gears and rolling-element bearings used in mining drivetrains, conveyors, hoists, and crushers [22]. The image dataset is therefore used to develop and benchmark the visual inspection method, not to represent mining components directly.

The methodology builds on established ML methods: Random Forest as a strong nonlinear ensemble baseline [23], XGBoost and LightGBM for gradient-boosted tree modelling [24], [25], Isolation Forest for anomaly detection [26], SHAP for explaining feature contributions [27], and Grad-CAM for interpreting CNN image predictions [28]. Python-based open-source libraries such as OpenCV, scikit-learn, XGBoost, and LightGBM provide practical implementation support [29]–[33].

III. METHODOLOGY

A. Data Collection

The project will use a layered dataset strategy: one primary real mining-machine sensor dataset, one clean industrial image dataset for computer vision, and two supporting heavy-equipment datasets for comparison or backup if data preparation challenges occur.

TABLE I
PROPOSED DATASETS AND ROLES IN THE CAPSTONE PROJECT

Dataset	Type	Relevance	Planned Use	Status
TiNA - Time-series Industrial Anomaly dataset	Multi-sensor time-series + alarms + maintenance codes	Real ArcelorMittal mining machine	Primary model for anomaly detection and maintenance-risk prediction	Primary
Industrial Machine Tool Component Surface Defect Dataset	RGB defect images	Surface pitting, a contact-fatigue failure also found in mining gears and bearings; used to benchmark the inspection method	OpenCV feature extraction, CNN classification, Grad-CAM	Primary CV
Scania Component X	Heavy-duty vehicle time-series + repair/time-to-event	Real heavy-equipment fleet maintenance benchmark	Optional validation/fallback for sensor predictive maintenance	Supporting
Scania APS Failure	Tabular heavy-truck failure data	Real imbalanced equipment failure problem with cost-sensitive evaluation	Optional benchmark for imbalance and maintenance cost analysis	Supporting

B. Data Preprocessing

For TiNA, the preprocessing pipeline will parse timestamps, sort observations in time order, audit missing values, remove duplicate records, identify constant or near-constant features, encode categorical fields, and build manageable time windows. Since TiNA is very large, the first prototype will use a selected time range that preserves all maintenance and alarm events within that range. The final subset will be expanded only if computation remains feasible.

TiNA features are anonymized, so the project will avoid unsupported physical interpretations such as claiming FEATURE1 is a specific pressure or temperature sensor. Instead, features will be treated as normalized industrial sensor streams. Maintenance and alarm fields will be used carefully to define prediction labels and event histories without leaking future information.

For the computer-vision dataset, preprocessing will include image integrity checks, label validation, resizing, normalization, train/validation/test partitioning, duplicate/near-duplicate checking where feasible, and OpenCV enhancement such as denoising, contrast adjustment, edge detection, and texture feature extraction.

C. Feature Engineering

Sensor features will be generated from time windows rather than single isolated rows. Planned features include rolling means, rolling standard deviations, rolling minima/maxima, slopes, lagged values, rate of change, event-history features, alarm-frequency features, and time since previous alarm or maintenance. The prediction target will be defined as maintenance or alarm risk within future horizons such as 1, 3, or 6 hours, selected after analyzing event frequency.

Computer-vision features will compare traditional OpenCV representations against learned CNN representations. Classical features may include Canny edges, contour statistics, histogram features, local texture descriptors, HOG/LBP-style features, defect-area proxies, and image intensity statistics. Deep learning will use a compact CNN or transfer-learning model, such as a pretrained ResNet [34], if computation allows.

D. Model Development

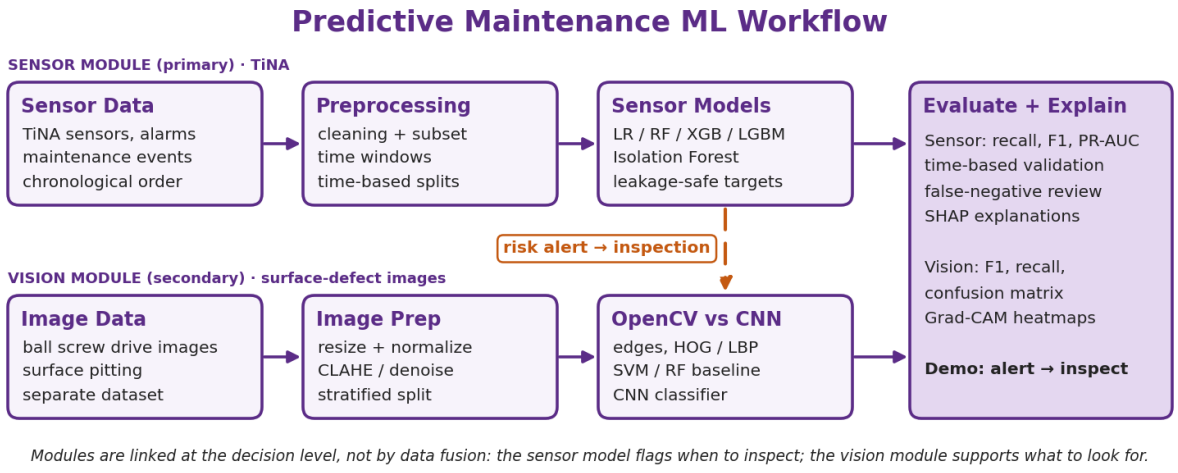


Fig. 1. Proposed multi-sensor and computer-vision predictive-maintenance workflow.

TABLE II
MODEL-DEVELOPMENT PLAN AND EVALUATION FOCUS

Task	Models	Main Metrics	Expected Output
Maintenance-risk prediction	Logistic Regression, Random Forest, XGBoost, LightGBM	Recall, F1, PR-AUC, ROC-AUC	Maintenance soon / no maintenance soon
Alarm-risk prediction	Random Forest, XGBoost, LightGBM	Recall, precision, F1, PR-AUC	Alarm probability in next horizon
Anomaly detection	Isolation Forest, One-Class SVM, Autoencoder if feasible	Anomaly score, precision/recall if labels available	Normal / abnormal behaviour
Visual defect classification	OpenCV + SVM/RF, CNN, transfer learning if feasible	Accuracy, F1, recall, confusion matrix	Defective / non-defective component surface
Explainability	SHAP, feature importance, Grad-CAM	Interpretability outputs	Top risk factors and image heatmaps

The two modules are integrated at the decision level rather than through data fusion. The sensor model identifies periods of elevated maintenance risk and explains the main contributing features with SHAP, while the vision module represents the inspection step that would follow an alert, classifying a component surface image and highlighting the suspected defect region with Grad-CAM. Because no public dataset pairs sensor histories with images from the same equipment, the modules are trained and evaluated on separate datasets, and the demonstration interface will show this alert-then-inspect workflow using representative samples. Data-level multimodal fusion remains future work.

E. Evaluation and Validation

- Use temporal validation for TiNA; do not randomly mix future records into training data.
- Create fixed train/validation/test periods or rolling time splits depending on the selected TiNA subset.
- Prioritize recall, F1-score, PR-AUC, false-negative rate, and false-positive trade-offs for rare maintenance events.
- Use accuracy only as a secondary metric because imbalanced maintenance data can produce misleading accuracy.
- Apply class weighting and training-only resampling (e.g., SMOTE [35]) where needed; no oversampling will occur before splitting.
- For images, maintain a fixed test set and avoid duplicate/near-duplicate leakage across partitions.
- Report both predictive performance and computational feasibility because the project must be completed on student hardware.

F. Ethical Considerations

Only public datasets will be used, and dataset licenses and citations will be respected. Proprietary Saskatchewan mine data will not be fabricated or represented as locally collected data. Because the project is an academic prototype, it will not be presented as a safety-certified or production-ready maintenance system. The proposal will also clearly communicate the limitations of anonymized features, class imbalance, possible dataset shift, and model uncertainty. In addition, the sensor and vision modules are trained on data from different equipment, so the visual model would require retraining and validation on real mining component images before any practical use.

G. Feasibility and Innovation

The project is feasible within 12 weeks because the scope is deliberately limited to two complementary modules: a primary sensor-based ML pipeline and a secondary visual-defect classification pipeline. Remaining Useful Life modelling, advanced deep sequence models, and full multimodal fusion are treated as stretch goals.

The innovation is an explainable alert-then-inspect predictive-maintenance workflow: leakage-safe event prediction and anomaly detection on real mining-machine sensor data identify when equipment is at risk, and OpenCV/CNN visual inspection with Grad-CAM supports what to look for once an alert is raised. The dashboard/demo remains secondary to ML experimentation.

H. Twelve-Week Work Plan and Weekly Reporting

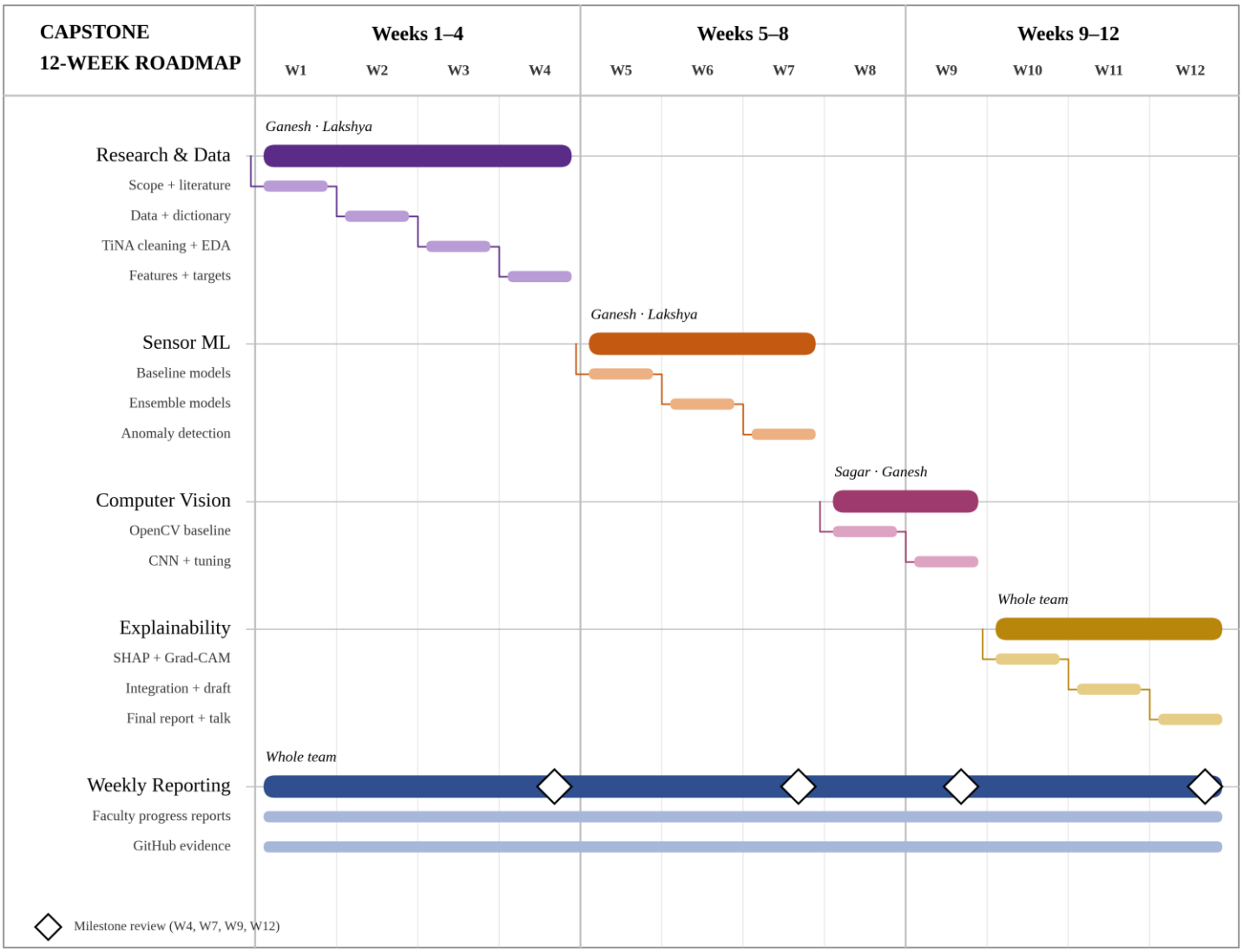


Fig. 2. Twelve-week capstone roadmap.

TABLE III

TWELVE-WEEK WORK PLAN WITH MEASURABLE WEEKLY REPORTING

Week	Main Focus	Instructor Evidence	Progress	Lead
1	Finalize title, scope, research questions, literature plan	Approved topic, project charter, initial references	7%	Ganesh
2	Dataset acquisition and data dictionary	Dataset summary, TiNA explanation, CV dataset explanation	14%	Lakshya

Week	Main Focus	Instructor Evidence	Progress	Lead
3	TiNA cleaning and EDA	Missing-value audit, event distributions, sensor plots	22%	Lakshya
4	Feature engineering	Rolling features, event targets, image preprocessing prototype	30%	Ganesh/Lakshya
5	Baseline ML models	Initial Logistic Regression/Decision Tree/RF results	40%	Ganesh
6	Advanced ensemble models	XGBoost/LightGBM comparison table	50%	Ganesh
7	Anomaly detection	Isolation Forest/One-Class SVM anomaly results	59%	Lakshya
8	OpenCV/CV baseline	Image preprocessing, OpenCV features, baseline classifier	68%	Sagar
9	CNN/vision improvement and model tuning	CNN results, hyperparameter tuning, imbalance handling	77%	Sagar/Ganesh
10	Explainability	SHAP plots, feature importance, Grad-CAM examples	86%	All
11	Integration, testing, draft report	Reproducible notebook/code, demo output, draft report	94%	All
12	Final report and presentation	Final metrics, presentation, GitHub repository, Q&A prep	100%	All

Weekly reporting format: planned work, completed work, data/ML experiments, quantitative results, challenges, GitHub evidence, next-week plan, and overall completion percentage.

I. Group Responsibilities

TABLE IV

PROPOSED GROUP RESPONSIBILITY SPLIT

Member	Primary Responsibilities	Shared Responsibilities
Ganesh Mishra	Coordination, sensor ML modelling, evaluation, report integration	Understand full pipeline, contribute to presentation and Q&A
Lakshya Sharma	Data cleaning, EDA, feature engineering, anomaly detection	Understand datasets, validation choices, and limitations
Sagar Patel	Computer vision, OpenCV/CNN experiments, visual results, testing support	Understand ML metrics, CV pipeline, and final demo

IV. EXPECTED RESULTS AND ANALYSIS

As this is a proposal, final numerical results are not yet available. Expected outputs include a cleaned TiNA working subset, a reproducible preprocessing pipeline, engineered time-window features, a maintenance-risk classifier, an anomaly-detection model, an OpenCV/classical-ML visual baseline, a CNN-based visual classifier, SHAP explanations, Grad-CAM heatmaps, and a concise dashboard/demo that shows equipment-health risk.

The final analysis will compare model families, feature sets, validation strategies, and error types. The project will report not only which model performs best, but also why the model is useful or limited for predictive maintenance. Special attention will be given to false negatives because missing a real degradation/failure event is more serious than raising an extra inspection alert.

V. CONCLUSION

The proposed capstone keeps predictive maintenance for mining equipment as the central research problem while using realistic public industrial data. TiNA provides the real mining-machine sensor foundation; the industrial image dataset provides a controlled computer-vision module; and supporting Scania datasets provide backup and context for heavy-equipment predictive maintenance. The two modules are linked at the decision level, reflecting how condition monitoring and inspection work together in maintenance practice, while their separate data sources are stated openly as a limitation. The proposal is feasible in 12 weeks and technically focused on core AI/ML.

VI. SCOPE OF FUTURE WORK

Future work could extend the prototype with Remaining Useful Life prediction, LSTM [36], GRU, or temporal-convolution models, advanced multimodal fusion between sensor and image streams collected from the same equipment, domain adaptation to Saskatchewan mining fleets, and validation using proprietary mine-maintenance records under appropriate permissions and confidentiality agreements.

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